Multimodal AI Teaching Copilot: Real-Time Slide Generation from Live Lecture Speech

Prachi Potdar, Paritosh Raut, Deepali Magdum, Swati Kale, and Sunita Patnaik
Department of Electronics and Telecommunication Engineering
JSPM's Rajarshi Shahu College of Engineering, Pune, Maharashtra, India – 411033
prachipotdar50@gmail.com, paritosh21raut@gmail.com, dipalimagdum@gmail.com, swatikale@gmail.com, sunitapatnaik@gmail.com

***Abstract*—Slides prepared before class cannot show what a teacher adds while explaining, live captions give students only verbatim text, and a large language model (LLM) that regenerates slides from a growing transcript rewrites what students are reading. This paper presents a teaching copilot that generates a structured, continuity-aware slide deck on the classroom projector from the teacher's live speech during the lecture. Locally transcribed speech is sent in bounded units to an LLM that returns a schema-validated interpretation. The novelty is that a deterministic planner, not the LLM, decides whether the live slide is updated, continued or replaced, so the deck stays stable, while transcript-grounding guards keep taught facts on screen. Formulas, layouts and open images are composed automatically, the teacher controls the deck live, and notes, assignments, quizzes, tests and decks are generated on request, checked against the teacher's own notes. On 36 held-out lectures (409 min; six speakers and six public MIT OCW and NPTEL lectures) on a laptop with a 4 GB GPU, slide changes matched annotated topic boundaries with precision 0.78, 86.6 % of key facts reached a slide, 0.9 % of statements were unsupported by the speech, and five teachers rated the slides 3.9–4.2 of 5. Median latencies were 0.68 s for speech recognition, 1.96 s per LLM call and 11.2 s from speech to slide, mostly a deliberate wait for a complete idea. Against an LLM-controlled baseline, rewrites of shown statements fell from 56 % to 3 %.**

***Index Terms*—Automatic slide generation, educational technology, large language models, real-time systems, speech recognition, human-in-the-loop**

I. Introduction

Slides prepared before a lecture fix its order in advance, so the examples and answers a teacher adds while explaining never reach the screen. Learners understand better from words with organized pictures than from words alone [1], if the visuals add no extraneous load [2], [3], but no teacher can author such visuals while teaching.

Existing tools solve only parts of this problem: live captions show verbatim text without the structure of the lesson [4]; slide generators work offline from a finished document or prompt [5]–[11]; and a large language model (LLM) [12], [13] applied to a growing transcript rewrites slides while students read them and adds untaught content [14]. Three problems remain open: (P1) projected material does not follow what is taught; (P2) live speech is not turned into a structured visual lesson; and (P3) LLM output is neither stable nor faithful to the teacher's words.

This paper presents the Multimodal AI Teaching Copilot, an implemented system that addresses all three. While the teacher speaks, the speech is transcribed locally, topic and subtopic are tracked, and a slide deck is generated on the projector. The novelty has three parts: (1) live slide generation from speech, built incrementally during the lecture instead of from a finished document; (2) a deterministic planner that gates the LLM, which only interprets bounded, schema-validated units, so every slide decision is a testable rule; and (3) continuity: a new concept extends the live slide, a topic change must be confirmed, and transcript-grounding guards keep taught content on screen. Teacher control, open images, study materials, a live quiz and test, and the teacher's own notes as study material complete the system, which is evaluated on 36 held-out lectures (409 min) against annotations, teacher ratings and an LLM-controlled baseline.

II. Related Work

A. Multimedia Learning and Slide Design

Mayer showed that words with pictures are understood better than words alone [1]; Sweller's cognitive load theory adds that working memory is limited [2], Mayer and Moreno listed ways to reduce extraneous load [3], and Bartsch and Cobern found that unrelated pictures lowered performance in PowerPoint lectures [15]. Hence the slides use short typed layouts and an image only for a concrete object that was taught.

B. Lecture Speech Recognition and Captioning

Whisper by Radford *et al.* [16], run with 8-bit weights by faster-whisper [17], and Silero VAD [18] form the speech front end. Kushalnagar *et al.* showed that verbatim real-time captions are hard for students to follow [4], and Gonzalez *et al.* found that LLM summaries of recorded lectures improved quiz scores [19]; both produce text, not a structured visual lesson (P2).

C. Topic Segmentation and Semantic Similarity

TextTiling by Hearst places subtopic boundaries where the similarity of adjacent blocks drops [20]; Sentence-BERT by Reimers and Gurevych compares sentence embeddings by cosine similarity [21], and Wang *et al.* distilled it into MiniLM [22]. These methods segment a complete transcript offline; the system applies the similarity drop online as one signal for a topic change and is evaluated with the boundary measure of Beeferman *et al.* [23].

D. Automatic Slide Generation

PPSGen selects sentences of a paper for slides by integer linear programming [7], D2S summarizes document text for each slide title [6], and DOC2PPT generates slides with text

TABLE I
COMPARISON OF THE PROPOSED SYSTEM WITH PRIOR SYSTEMS

System	Input	Produced	Topic structure	Faithfulness to source	Slide sequence
Live captions [4], [16]	Speech	Live	None (text stream)	Verbatim text	None
Lecture summaries [19]	Recorded lecture	Offline	None (summary)	LLM summary	None
PPSGen, D2S, DOC2PPT [5]–[7]	Paper / document	Offline	Document sections	Extracted from document	Whole deck at once
LLM slide generators [8]–[11]	Topic / document / prompt	Offline	LLM outline / planning	Retrieved sources [8], [9] or none	Whole deck at once
Proposed system	**Live speech**	**During lecture**	**Online, confirmed topic / subtopic**	**Transcript coverage and formula guards**	**Deterministic planner + teacher**

and figures from scientific documents [5]. With LLMs, Xie *et al.* ground educational slides in retrieved sources [8], SlideBot adds multi-agent planning [9], PPTAgent edits reference slides [10], and AutoPresent writes slides as code [11]. All of them build a whole deck offline from written input, whereas a lecture arrives incrementally as noisy speech and its slides must stay stable while students read them (P1, P3).

E. LLMs in Education and Faithful Generation

Kasneci *et al.* stressed teacher oversight of LLMs in education [12], so the teacher overrides every automatic decision. Ji *et al.* surveyed hallucination [14], and Willard and Louf constrained generation to a JSON schema [24], which fixes the format but not the facts; the output is therefore also checked against the transcript. Kurdi *et al.* reviewed question generation for education [25], the basis of the generated assignments, quizzes and tests.

Table I compares the systems on five properties that decide whether slides are usable in a live classroom. It shows why the problem is open: prior systems either stream unstructured text or build a whole deck at once from written input, and none decides incrementally which slide to change while students read. This is the gap addressed here; the last column is tested against a baseline in Section IV.

III. METHODOLOGY

A. System Overview

A lecture is processed as a stream through six stages (Fig. 1): speech capture and recognition; filtering and concept tracking; a gate that groups finished sentences into units; bounded LLM interpretation of each unit; a deterministic planner and composer that turn it into slide operations and layouts; and delivery to the projector, the teacher and students. Four rules guide the design: the LLM never controls the flow, it never sees the whole transcript, a new concept is not by itself a new slide, and user interfaces send commands instead of changing the state.

All components run in one Python *asyncio* process and exchange typed events on an in-process bus; one store owns the lecture state, and every event is logged in SQLite, from which each lecture is replayed for evaluation and materials. Recognition runs on the GPU, MiniLM and CLIP on the CPU (ONNX), and the LLM on the Groq free tier (gpt-oss-120b [13], then qwen3.8-27b when its quota is spent), as a 7B-class local model does not fit beside the speech model in 4 GB.

B. Speech Capture and Concept Tracking

Silero VAD [18] segments 16 kHz audio (utterances up to 15 s), and faster-whisper [17] with Whisper medium.en [16] (int8/float16 on CUDA, beam size 1, subject-keyword prompt) transcribes them. A deterministic filter removes classroom management, meta talk and fillers. Each content utterance is embedded with MiniLM [22]; with the L2-normalized embedding e_t and concept centroid c_{t-1}, the shift score is the cosine distance used with Sentence-BERT [21], and the centroid an exponentially weighted mean (proposed here):

$$s_t = 1 - \mathbf{e}_t^{\top}\mathbf{c}_{t-1} \quad (1)$$

$$\mathbf{c}_t = \mathrm{norm}((1-\alpha_t)\mathbf{c}_{t-1} + \alpha_t \mathbf{e}_t) \quad (2)$$

where $\mathrm{norm}(x) = x/\|x\|$ and $\alpha = 0.35$, raised to 0.6 at a boundary so the centroid re-seeds on the new concept. Following the similarity-drop criterion of TextTiling [20], a boundary is signalled when

$$b_t = \kappa_t \vee (s_t \geq \tau), \tau = 0.75 \quad (3)$$

where κ_t marks a cue phrase ("moving on to", "next"). Shift scores within one facet were 0.3–0.6, so τ was set high by hand; cue words carry most boundaries and the LLM decides the final relation.

C. Gate and Token Budget

Content accumulates in a buffer until a rule-based gate (proposed here) fires. With p_t the silence after a finished sentence, w_t the buffered words and a_t the age of the oldest pending line (s),

$$G_t = (p_t \geq 1.2 \wedge w_t \geq 6) \vee w_t \geq 40 \vee b_t \vee a_t \geq 12 \quad (4)$$

with an 8 s floor between calls; fragments under six words join the next unit. The floor follows from the free quota of 200,000 tokens per key and model per day (85 calls of about 2,300 tokens).

D. Bounded Interpretation and Transcript-Grounding Safeguards

The prompt holds the subject, level, an outline (≤ 150 tokens), the current slide with numbered items, a rolling summary (≤ 120 tokens) and the new lines, within 3,000 tokens (temperature 0.2). The LLM returns JSON with topic, subtopic, a relation (same concept, elaboration, sub-concept, sibling, new topic, digression), typed content (definition, facts, steps, comparison, timeline, formula, example, …), an image hint and *concerns* about possibly wrong statements. Pydantic validates it; after one failed repair, a deterministic

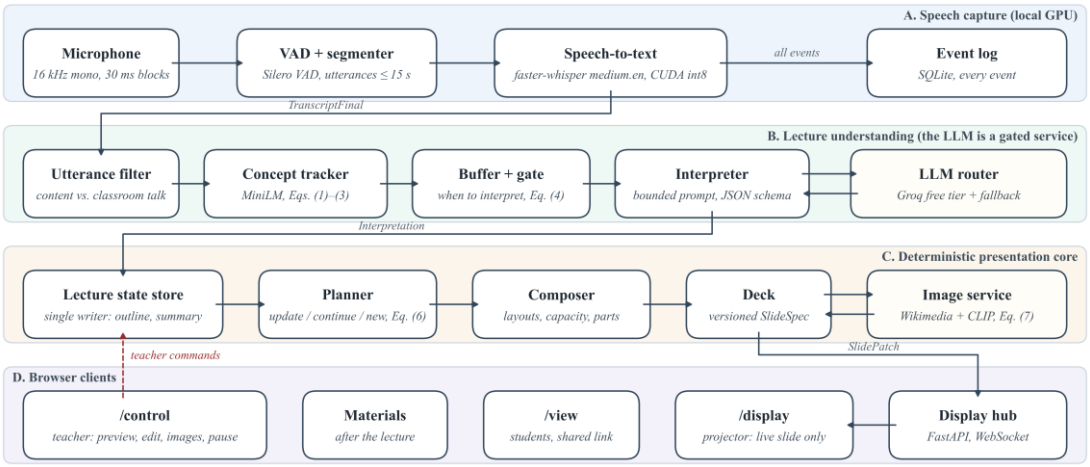


Fig. 1. Implemented architecture. Lanes A–C run in one Python process; the LLM is called only by the interpreter through the router. Browser clients receive versioned slide patches and send teacher commands back; the user interface never changes the core state directly.

fallback shows the spoken sentences as points. A coverage guard compares every content sentence u (≥ 6 words) with the output O; with $W(\cdot)$ the set of word stems and a recall-style overlap as in ROUGE-1 [26],

$$|W(u) \cap W(O)| \leq \frac{1}{3}|W(u)| \Rightarrow u \text{ is shown as said} \quad (5)$$

so a fact the model left out still reaches the slide. A formula guard flags changed formula tokens, and a correction is applied only above a confidence of 0.75 (factual) or 0.4 (mis-hearing); otherwise the slide shows what was said and the teacher gets a card with both versions. These checks keep slides faithful to the transcript; they do not prove a statement correct. The router rotates free keys under token buckets with a 15 s deadline per unit.

E. Slide Planner

The planner turns each interpretation into one operation: *update* the working slide, *continue* on a new slide of the same topic, open a *new* topic slide, *retitle*, or nothing (Fig. 2). A topic change must be confirmed (rule proposed here), because a single unit may be misread:

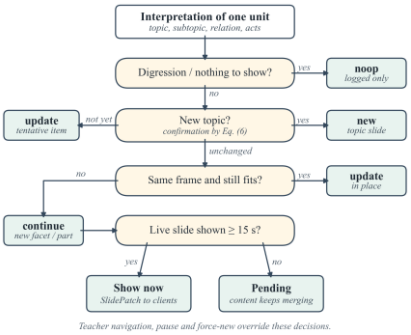


Fig. 2. Planner decision for one interpreted unit. The LLM supplies the relation and content; the decision itself is rule-based and unit-tested.

$$C_n = B_n \vee \left(\bar{s}_n^{\max} \geq \tau\right) \vee \left(T_n \equiv \tilde{T}_{n-1}\right) \quad (6)$$

where B_n is true if a boundary (3) occurred in unit n, $\bar{s}_n^{\max}$ is its largest shift score smoothed over two utterances, T_n its topic and $\tilde{T}_{n-1}$ an unconfirmed candidate from the previous unit. Unconfirmed content is shown tentatively and moved once confirmed. A minimum dwell of 15 s (6 s for the next part of a full slide) prevents rapid changes, near-duplicates (similarity ≥ 0.85) are merged, and teacher edits are never overwritten. All thresholds were set by hand on D1 (Section IV).

F. Composer and Image Retrieval

The composer maps items to definition cards, key points, fact tiles, process flows, comparison tables, timelines, cause-effect chains, classification trees and formulas; a height model of the 16:9 stage moves overflow to a further part (I, II, …) instead of shrinking text. Spoken mathematics is converted to LaTeX deterministically and typeset with KaTeX. For a concrete object the LLM suggests a query q; openly licensed files from Wikipedia and Wikimedia Commons (source, author and licence kept) are scored by CLIP's zero-shot image–text similarity σ [27], and image I is accepted when

$$\sigma(q,I) \geq 0.22 \wedge \sigma(q,I) - \max_{d \in D}\sigma(d,I) \geq 0.01 \quad (7)$$

where D holds generic distractor prompts ("a logo", "a map"); labelled diagrams rank first, and an image is placed only if query and slide text have a MiniLM similarity of at least 0.25.

G. Live Display and Teacher Control

A FastAPI server pushes versioned slide patches over WebSocket to /display (projector), /control (teacher) and /view (students, shared link). The control view (Fig. 3) offers an editable preview, navigation, pause, blank screen, editing, image search and upload, mistake cards, the lecture structure tree and themes.

H. Lecture Materials, Quiz, Test and Study Material

On request, the lecture, past lectures or an archive chapter become a summary slide, study notes and an assignment (PDF) and a PPTX deck (Fig. 4). A quiz (timed multiple choice, leaderboard on the projector) and a test (set duration, scores to the teacher) are generated from the taught content, edited by the teacher and answered through the shared link, once per roll number. The teacher's own notes are indexed

locally as study material: a clearly matching passage (MiniLM similarity ≥ 0.5, ≤ 150 tokens) enters the prompt only to name and spell terms, a contradiction becomes a concern citing file and page, its figures precede web images, its rare terms extend the recognizer prompt, and quiz answers are checked against it.

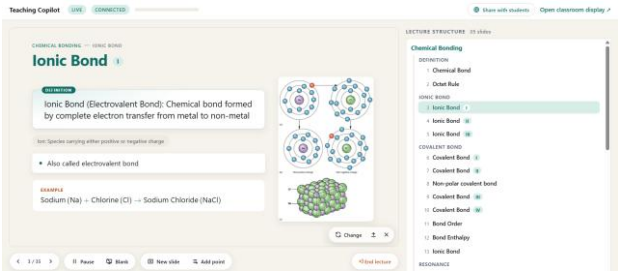
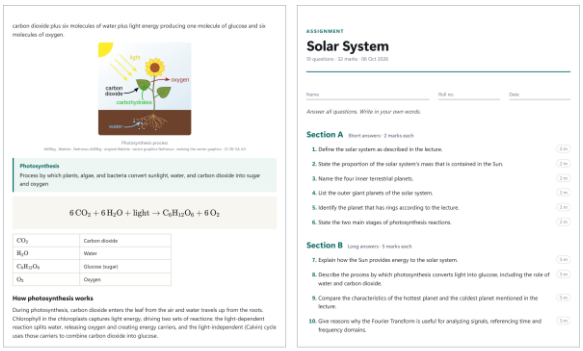


Fig. 3. Teacher control view in a live chemistry lecture: editable preview of the projected slide with image tools and lecture controls (left) and the lecture structure tree built by the system, topic → facet → slides (right).



(a) Study notes (PDF, page 3) (b) Assignment (PDF)

Fig. 4. Materials generated on request with the real LLM: (a) study notes with a retrieved diagram, definition, typeset equation and legend; (b) an assignment grounded in what was taught.

IV. Experimental Evaluation

A. Setup and Data

All runs used a laptop with an RTX 3050 GPU (4 GB) and 16 GB RAM. All thresholds were set on D1, six lectures spoken by the authors (114.1 min). The held-out set D2 (Table II) has 30 lectures spoken live by six speakers into the laptop microphone (Recorded, physics to economics) and six 10-min public MIT OpenCourseWare and NPTEL lectures played in as audio in real time (Public). Of its 1,619 units, 72 (4.4 %) fell back because a quota was spent or the provider failed, and still produced slides. Slides were not edited during a test; only two items were deleted in all of D2.

TABLE II
Evaluation Data and Latency (Session Logs)

Set	Lect.	Min	Units (fb)	Slides	STT	LLM	E2E (p95)
D1	6	114.1	389 (45)	163	0.66	1.35	12.5 (36.8)
D2 Recorded	30	349.2	1275 (59)	453	0.71	1.92	10.1 (23.2)
D2 Public	6	60.0	344 (13)	52	0.60	2.23	17.1 (93.4)
D2 all	**36**	**409.2**	**1619 (72)**	**505**	**0.68**	**1.96**	**11.2 (53.0)**

fb: fallback units; STT, LLM: median per utterance / call (s); E2E: median (95th percentile) from the end of a spoken line to its display (s); Public includes web4, web6 (provider outages).

Annotation. Two authors (A and B) independently annotated 18 D2 lectures (12 Recorded, 6 Public; 191 min). Before seeing the slides, each marked topic boundaries (where a teacher would start a new board) and key facts in the transcript, then judged whether each key fact was on a slide and whether each slide statement was said and correct. A system boundary is the utterance that opened a new content slide. Boundaries are scored by precision, recall and F1 within ±1 utterance and by P_k [23], the probability that two utterances half a segment apart are wrongly split or joined; agreement is Cohen's $\kappa = (p_o - p_e)/(1 - p_e)$ [28] on the scale of Landis and Koch [29]. *Ratings.* Five teachers (T1–T5), not involved in development, rated 30 random D2 slides, each shown with what was said, from 1 to 5 on four criteria; their consistency is the ICC(2,1) of Shrout and Fleiss [30]. *Baseline.* On the same units of two D1 lectures, an LLM-controlled baseline gets its last slide and the last 60 s of transcript and returns the slide to show, without planner, guards or composer.

B. Topic Boundaries and Content Accuracy

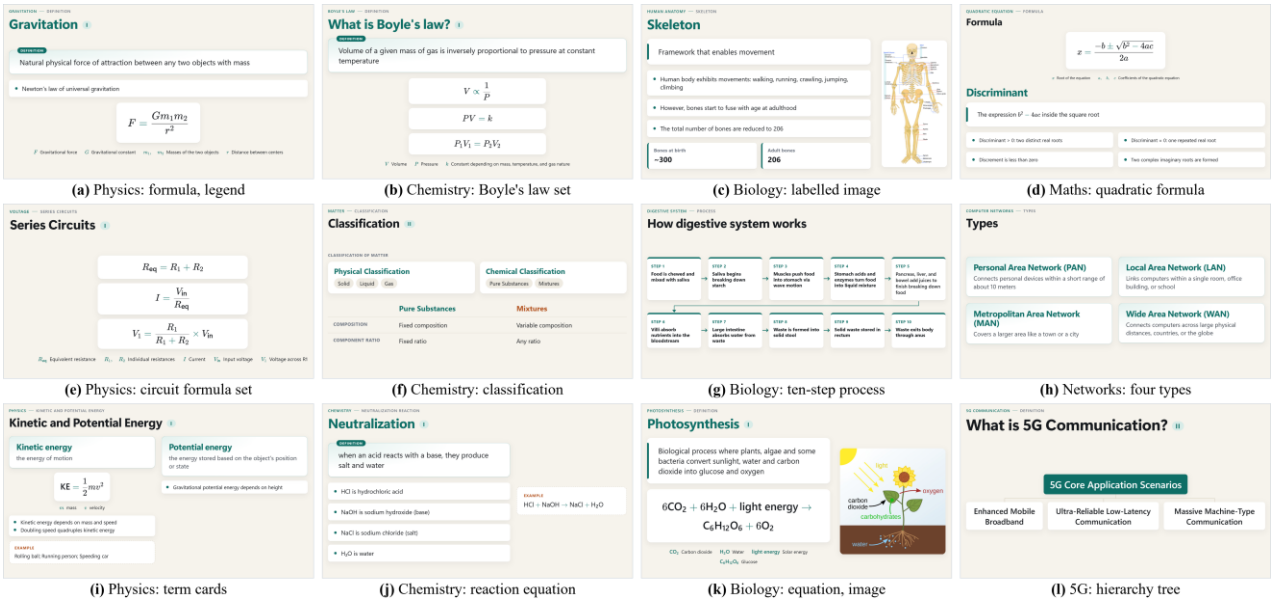
TABLE III
Annotated Accuracy and Teacher Ratings (D2)

Measure	Recorded	Public	All
Boundary precision / recall	0.81 / 0.54	0.73 / 0.19	0.78 / 0.42
Boundary F1 / P_k	0.59 / 0.29	0.29 / 0.37	0.49 / 0.32
Key-fact coverage	88.0 %	84.0 %	86.6 %
Unsupported / wrong	0.9 / 3.6 %	0.9 / 4.0 %	0.9 / 3.7 %
κ boundaries (agreement)	0.81 (94 %)	0.56 (90 %)	0.72 (92 %)
κ supported (agreement)	0.10 (90 %)	−0.01 (85 %)	0.07 (89 %)
Rating: correct. / relevance	4.20 / 3.94	4.06 / 3.80	4.15 / 3.89
Rating: readab. / usefulness	4.12 / 3.93	4.26 / 3.76	4.17 / 3.87

Means over annotators A and B; boundaries ±1 utterance; unsupported / wrong: share of slide statements; ratings: mean of T1–T5 (1–5; 20 Recorded, 10 Public slides).

Slides change where the topic changes (Table III): boundary precision is 0.78, so a projected slide is rarely replaced within a topic, and P_k is 0.32. Recall is 0.42 (F1 0.49), lower by design: a new concept is not a new slide, so the planner merges into the live slide the sub-points that annotators marked as boundaries, most often in the Public lectures (recall 0.19). For a projector a missed sub-boundary costs less than a false change, so precision and P_k matter most. Of 669 key facts, 86.6 % reached a slide (partly shown counts ½), and only 0.9 % of 2,504 statement judgments were not supported by the speech. Of the 3.7 % marked wrong, 41 of 79 were also marked as said (a slip or mis-hearing): slides follow the speech, and corrections go to the teacher's concern cards. Boundary agreement is substantial (κ = 0.72 [29]); for "supported" the annotators agreed on 88.8 %, but κ is 0.07 because nearly all statements were supported, which pushes chance agreement toward 1 (the prevalence paradox [31]).

Reported separately, test29, interpreted entirely by the fallback (quota spent), covered 83.1 % of key facts with 0.7 % unsupported statements but rarely opened a slide (recall 0.04); the 17 LLM lectures reach precision 0.77 and recall 0.45. The teachers rated the slides 3.87–4.17 (ICC(2,1) 0.45–0.56, moderate). In Fig. 5 the layout follows the kind of content, not the subject.



(a) Physics: formula, legend **(b)** Chemistry: Boyle's law set **(c)** Biology: labelled image **(d)** Maths: quadratic formula

(e) Physics: circuit formula set **(f)** Chemistry: classification **(g)** Biology: ten-step process **(h)** Networks: four types

(i) Physics: term cards **(j)** Chemistry: reaction equation **(k)** Biology: equation, image **(l)** 5G: hierarchy tree

Fig. 5. Projected slides from the six D1 lectures, one subject per column. (c), (d), (f) and (k) are screenshots of the running application; the others are recorded LLM interpretations re-rendered by the current display engine without new LLM calls. Slide text is as spoken, including the slip "Discrement" in (d).

C. End-to-End Latency

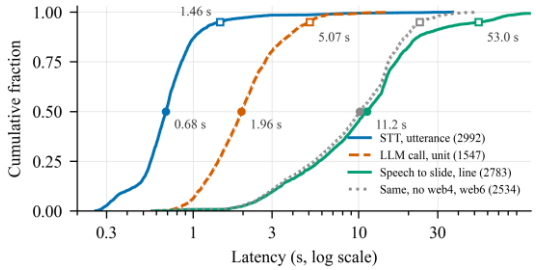


Fig. 6. Latency over D2 (log scale) per utterance, LLM call and spoken line; dotted: without the two lectures with provider outages. Filled: medians; open: 95th percentiles.

The models are fast (Fig. 6, Table II): over D2, speech-to-text takes a median of 0.68 s, an LLM call 1.96 s, and planning and composing 2 ms. From the end of a spoken line to its content on the display the median is 11.2 s, of which 7.6 s (median) is waiting in gate and queue, mostly the deliberate gate that waits for a complete idea so that a slide changes once per idea, not once per sentence. In web4 and web6 the free provider failed (32 and 6 calls: timeouts, HTTP 429), units queued, and the 95th percentile of D2 rose to 53.0 s; without these two lectures the median is 10.2 s and the 95th percentile 23.4 s (2,534 lines). The fallback kept slides coming during these outages.

D. Comparison With an LLM-Controlled Baseline

TABLE IV
PROPOSED SYSTEM VS. LLM-CONTROLLED BASELINE (SAME UNITS)

Lecture / system	Slides	Rewritten	Missing
Chemistry: proposed	36	8 / 317 (3 %)	5 / 136 (4 %)
Chemistry: baseline	15	226 / 405 (56 %)	220 / 293 (75 %)
Biology: proposed	11	0 / 74 (0 %)	0 / 39 (0 %)
Biology: baseline	13	48 / 160 (30 %)	44 / 90 (49 %)

Rewritten: statements of a shown slide that its next version rewrote or removed; missing: distinct statements shown but absent from the final deck.

On the same units, the baseline kept one slide for long stretches and rewrote it (Table IV): in chemistry it rewrote or removed 56 % of the statements students had already seen and lost 75 % from its final deck. The proposed system appends to the live slide and moves on, rewriting 3 % (mostly mis-hearing fixes) and losing 4 %; in biology the baseline rewrote 30 % and lost 49 %, the proposed system none. In D1, 33 of 63 image searches found an accepted image; on a 30-concept labelled set the first image was relevant for all 28 that received one.

V. DISCUSSION

When the planner, not the LLM, owns the slide sequence, a stable deck is built from live speech on modest hardware: the baseline lost most of what it had shown, whereas the proposed system changed slides mainly at topic changes and added almost nothing unsaid. The cost is a delay of about 11 s, set by the wait for a complete idea rather than by the models. All 553 unit and integration, 39 browser and 25 GPU tests of the implementation pass. Limitations: the free-tier provider caused outages, boundary recall is moderate, and without a classroom study, the next step, no claim is made about learning outcomes.

VI. CONCLUSION

This paper presented the Multimodal AI Teaching Copilot, which generates a continuity-aware slide deck from live lecture speech, with the LLM as a bounded interpreter and a deterministic planner in control of the slides. On 36 held-out lectures (409 min) it built 505 slides that changed at annotated topic boundaries with precision 0.78, covered 86.6 % of key facts with 0.9 % unsupported statements and were rated 3.9–4.2 of 5 by five teachers, and it rewrote 3 % instead of 56 % of the statements already shown, compared with an LLM that controls the slides.

REFERENCES

[1] R. E. Mayer, *Multimedia Learning*, 2nd ed. Cambridge, U.K.: Cambridge Univ. Press, 2009.

[2] J. Sweller, "Cognitive load during problem solving: Effects on learning," *Cognitive Sci.*, vol. 12, no. 2, pp. 257–285, 1988.

[3] R. E. Mayer and R. Moreno, "Nine ways to reduce cognitive load in multimedia learning," *Educ. Psychologist*, vol. 38, no. 1, pp. 43–52, 2003.

[4] R. S. Kushalnagar, W. S. Lasecki, and J. P. Bigham, "Accessibility evaluation of classroom captions," *ACM Trans. Accessible Comput.*, vol. 5, no. 3, Art. no. 7, 2014.

[5] T.-J. Fu, W. Y. Wang, D. McDuff, and Y. Song, "DOC2PPT: Automatic presentation slides generation from scientific documents," in *Proc. AAAI Conf. Artif. Intell.*, vol. 36, no. 1, 2022, pp. 634–642.

[6] E. Sun, Y. Hou, D. Wang, Y. Zhang, and N. X. R. Wang, "D2S: Document-to-slide generation via query-based text summarization," in *Proc. NAACL-HLT*, 2021, pp. 1405–1418.

[7] Y. Hu and X. Wan, "PPSGen: Learning-based presentation slides generation for academic papers," *IEEE Trans. Knowl. Data Eng.*, vol. 27, no. 4, pp. 1085–1097, Apr. 2015.

[8] E. Xie, G. Xiong, H. Yang, O. Coleman, M. Kennedy, and A. Zhang, "Leveraging grounded large language models to automate educational presentation generation," in *Proc. Large Found. Models Educ. Assessment*, PMLR, vol. 264, 2025, pp. 207–220.

[9] E. Xie, D. Waterfield, M. Kennedy, and A. Zhang, "SlideBot: A multi-agent framework for generating informative, reliable, multi-modal presentations," in *Proc. AAAI Conf. Artif. Intell.*, vol. 40, no. 48, 2026, pp. 40907–40915.

[10] H. Zheng *et al.*, "PPTAgent: Generating and evaluating presentations beyond text-to-slides," in *Proc. EMNLP*, 2025, pp. 14402–14418.

[11] J. Ge *et al.*, "AutoPresent: Designing structured visuals from scratch," in *Proc. IEEE/CVF CVPR*, 2025, pp. 2902–2911.

[12] E. Kasneci *et al.*, "ChatGPT for good? On opportunities and challenges of large language models for education," *Learn. Individual Differences*, vol. 103, Art. no. 102274, 2023.

[13] OpenAI, "gpt-oss-120b & gpt-oss-20b model card," arXiv:2508.10925, 2025.

[14] Z. Ji *et al.*, "Survey of hallucination in natural language generation," *ACM Comput. Surv.*, vol. 55, no. 12, Art. no. 248, 2023.

[15] R. A. Bartsch and K. M. Cobern, "Effectiveness of PowerPoint presentations in lectures," *Comput. Educ.*, vol. 41, no. 1, pp. 77–86, 2003.

[16] A. Radford, J. W. Kim, T. Xu, G. Brockman, C. McLeavey, and I. Sutskever, "Robust speech recognition via large-scale weak supervision," in *Proc. ICML*, 2023, pp. 28492–28518.

[17] SYSTRAN, "faster-whisper: Faster Whisper transcription with CTranslate2," GitHub. [Online]. Available: https://github.com/SYSTRAN/faster-whisper

[18] Silero Team, "Silero VAD: Pre-trained enterprise-grade voice activity detector," GitHub, 2021. [Online]. Available: https://github.com/snakers4/silero-vad

[19] H. Gonzalez *et al.*, "Automatically generated summaries of video lectures may enhance students' learning experience," in *Proc. 18th Workshop BEA*, 2023, pp. 382–393.

[20] M. A. Hearst, "TextTiling: Segmenting text into multi-paragraph subtopic passages," *Comput. Linguistics*, vol. 23, no. 1, pp. 33–64, 1997.

[21] N. Reimers and I. Gurevych, "Sentence-BERT: Sentence embeddings using Siamese BERT-networks," in *Proc. EMNLP-IJCNLP*, 2019, pp. 3982–3992.

[22] W. Wang, F. Wei, L. Dong, H. Bao, N. Yang, and M. Zhou, "MiniLM: Deep self-attention distillation for task-agnostic compression of pre-trained transformers," in *Proc. NeurIPS*, vol. 33, 2020, pp. 5776–5788.

[23] D. Beeferman, A. Berger, and J. Lafferty, "Statistical models for text segmentation," *Mach. Learn.*, vol. 34, no. 1–3, pp. 177–210, 1999.

[24] B. T. Willard and R. Louf, "Efficient guided generation for large language models," arXiv:2307.09702, 2023.

[25] G. Kurdi, J. Leo, B. Parsia, U. Sattler, and S. Al-Emari, "A systematic review of automatic question generation for educational purposes," *Int. J. Artif. Intell. Educ.*, vol. 30, no. 1, pp. 121–204, 2020.

[26] C.-Y. Lin, "ROUGE: A package for automatic evaluation of summaries," in *Proc. ACL Workshop Text Summarization*, 2004, pp. 74–81.

[27] A. Radford *et al.*, "Learning transferable visual models from natural language supervision," in *Proc. ICML*, 2021, pp. 8748–8763.

[28] J. Cohen, "A coefficient of agreement for nominal scales," *Educ. Psychol. Meas.*, vol. 20, no. 1, pp. 37–46, 1960.

[29] J. R. Landis and G. G. Koch, "The measurement of observer agreement for categorical data," *Biometrics*, vol. 33, no. 1, pp. 159–174, 1977.

[30] P. E. Shrout and J. L. Fleiss, "Intraclass correlations: Uses in assessing rater reliability," *Psychol. Bull.*, vol. 86, no. 2, pp. 420–428, 1979.

[31] A. R. Feinstein and D. V. Cicchetti, "High agreement but low kappa: I. The problems of two paradoxes," *J. Clin. Epidemiol.*, vol. 43, no. 6, pp. 543–549, 1990.